INCH-POUND

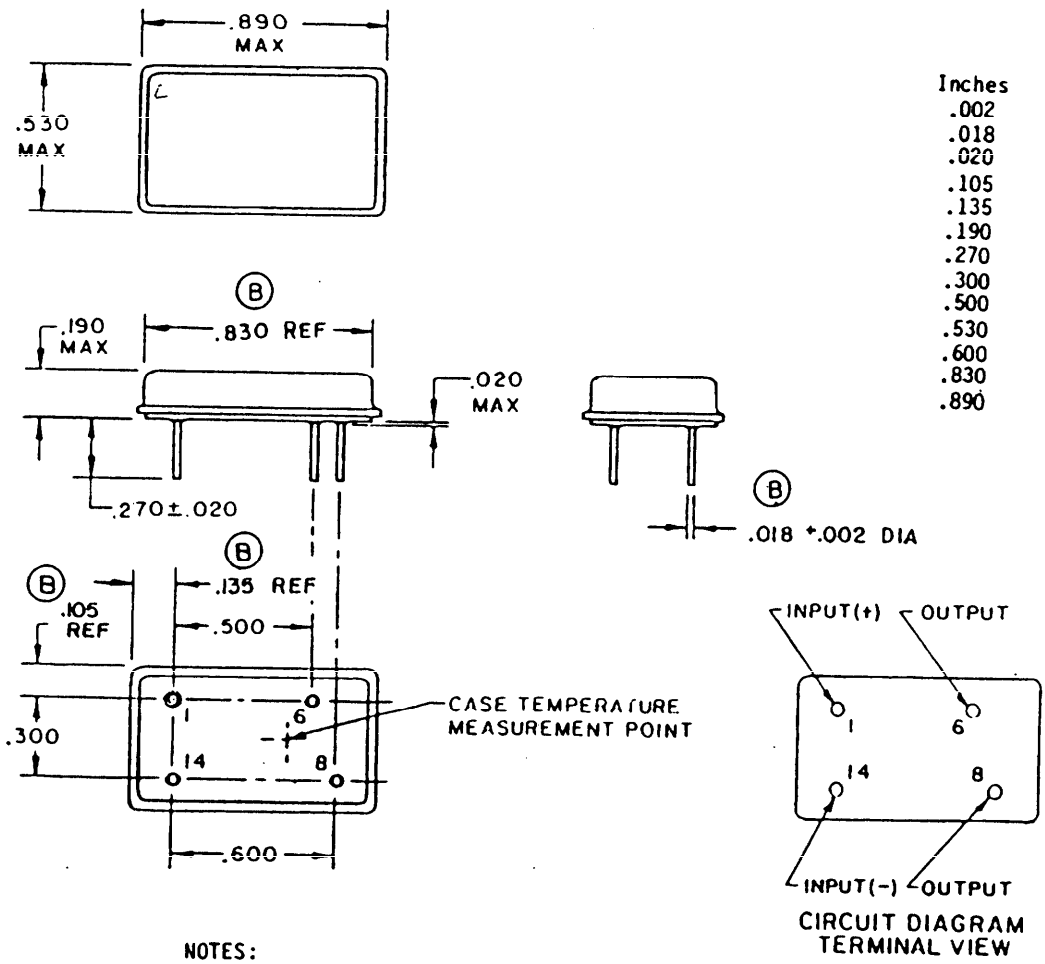
MIL-R-28750/98
14 November 1988
SUPERSEDING
MIL-R-28750/9A(EC)
8 April 1982

MILITARY SPECIFICATION SHEET

RELAY, SOLID STATE, HERMETICALLY SEALED, OPTICALLY ISOLATED, ZERO VOLTAGE TURN ON, 1 AMPERE (2 AMPERES WITH HEAT SINK), 40 - 440 HZ, POWER SWITCHING, SPST (N.O.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-28750.



Inches	mm
.002	0.05
.018	0.46
.020	0.51
.105	2.67
.135	3.43
.190	4.83
.270	6.86
.300	7.62
.500	12.70
.530	13.46
.600	15.24
.830	21.08
.890	22.61

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only.

FIGURE 1. Outline drawing and dimensions.

Ⓑ denotes changes

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REQUIREMENTS:

INPUT DATA:

Input voltage range: 3.8 V dc to 32 V dc.

Ⓑ Assured turn-on voltage: $\leq$3.8 V dc.

Ⓑ Assured turn-off voltage: $\geq$1.5 V dc.

Input current: 18 mA dc maximum.

Turn-on time: 1/2 cycle ac Hz.

Turn-off time: 1 cycle ac Hz.

Bias voltage: Not applicable.

OUTPUT DATA:

Rated output current: See figures 2 and 3.

Rated output voltage: 250 V ac rms maximum, 40 to 440 Hz.

Output voltage drop: 1.5 V ac maximum.

Output leakage current: 3 mA rms maximum.

Crosstalk: Not applicable.

Transient voltage: 130 V rms at 400 Hz, diode peak inverse voltage 500 V peak (5 seconds maximum). (Relay can be externally protected to meet the requirements of MIL-STD-704, category B.)

Electrical system spike: Not applicable.

Overload: 1.5 times the rated current (no heat sink), 10 percent duty cycle (see figure 4).

DC offset voltage: ±100 millivolts maximum.

Waveform distortion: 4 V rms maximum, initial turn on: ±10 V peak maximum.

Minimum current: Applicable (load current shall be varied between 10 and 100 percent of rated load).

ELECTRICAL DATA:

Dielectric withstanding voltage: 1,500 V ac rms, 60 Hz.

Insulation resistance: 100 megohms minimum.

Isolation: 10 picofarads maximum.

Power dissipation: 2.00 watts maximum (no heat sink).

ENVIRONMENTAL DATA:

Temperature:

Operating: -55°C to +110°C.

Storage: -55°C to +125°C.

Thermal shock: Applicable.

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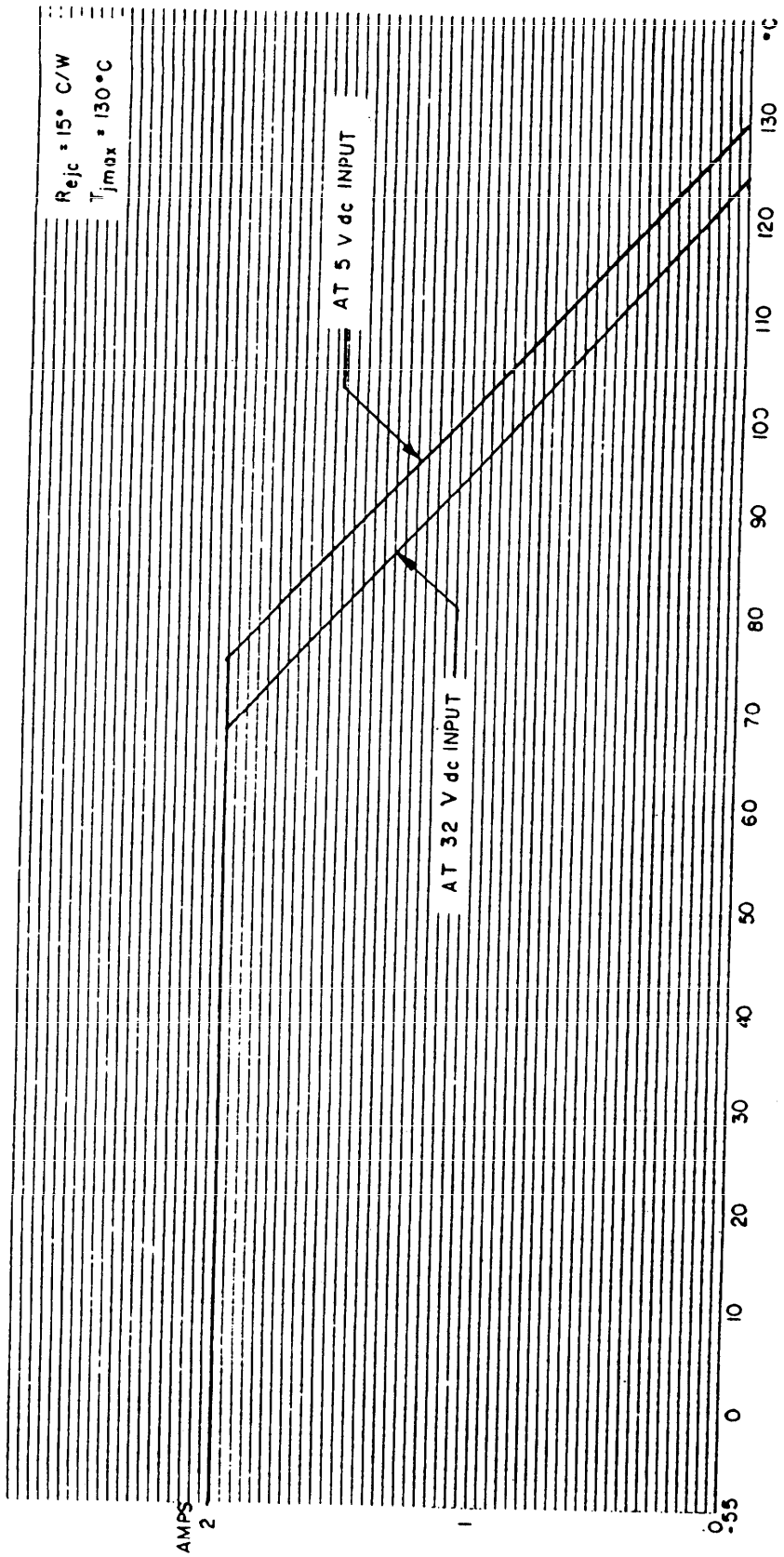


FIGURE 2. Maximum load current vs. case temperature.

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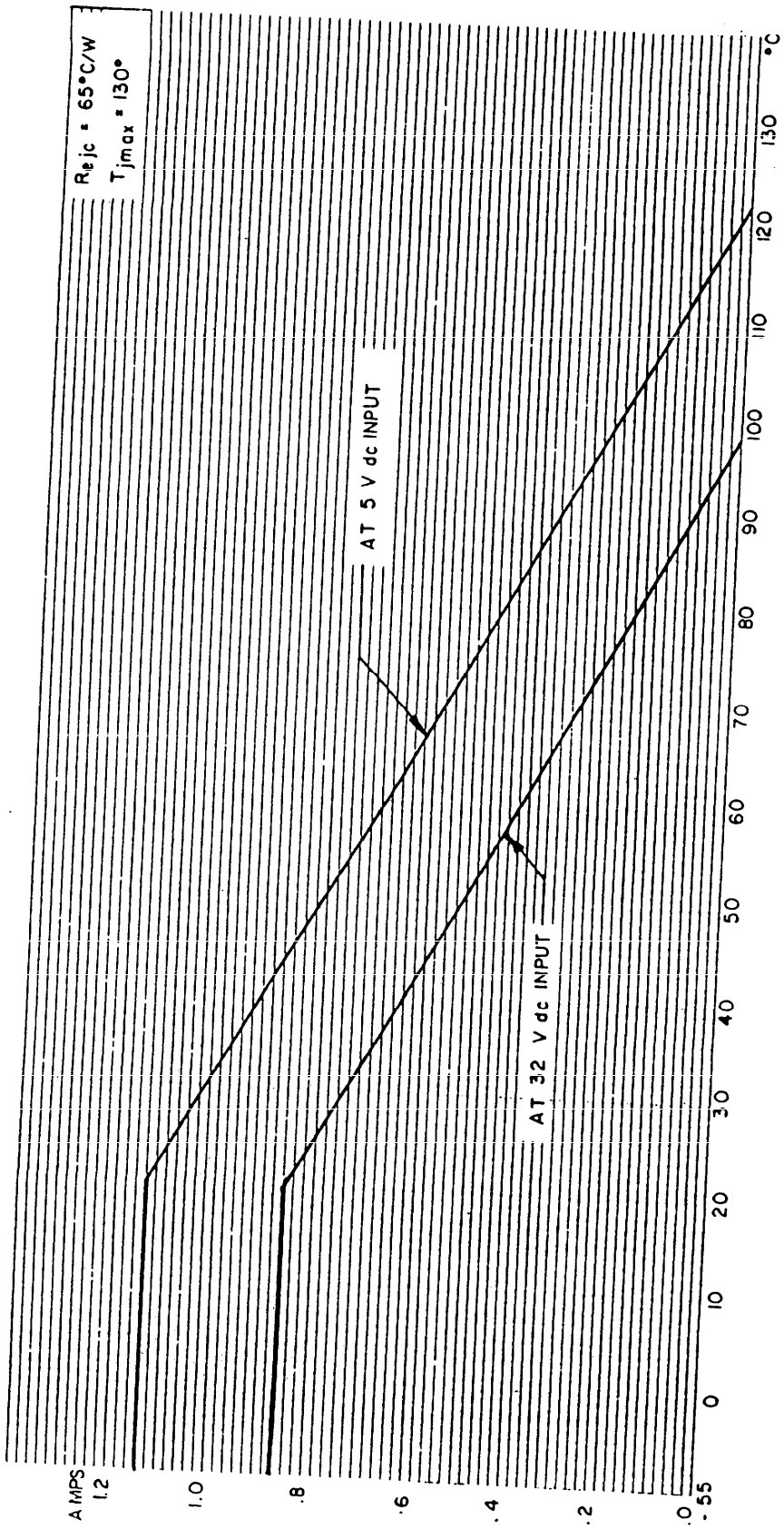


FIGURE 3. Maximum load current vs. ambient temperature.

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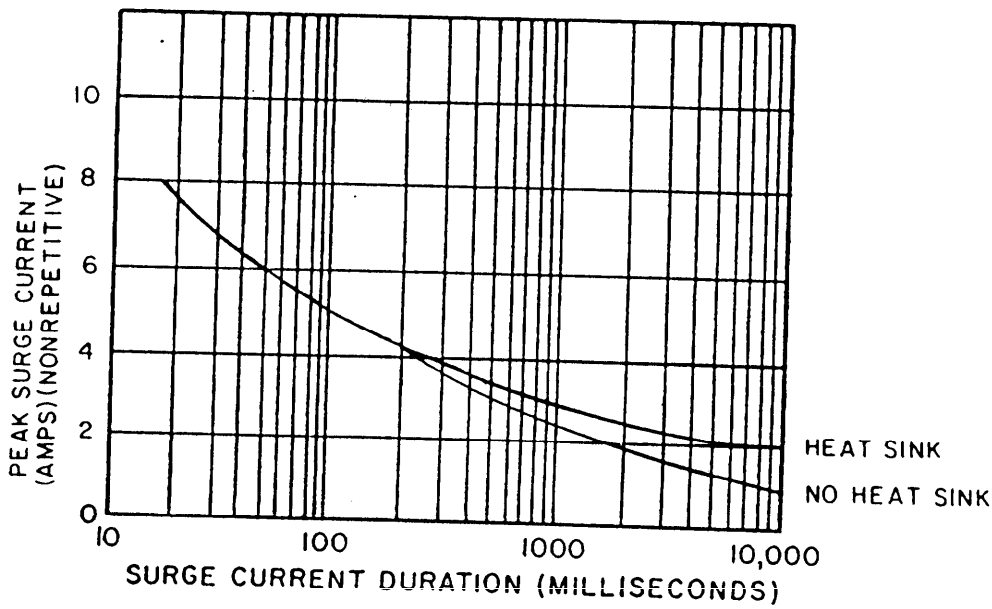


FIGURE 4. Peak surge current vs. surge current duration.

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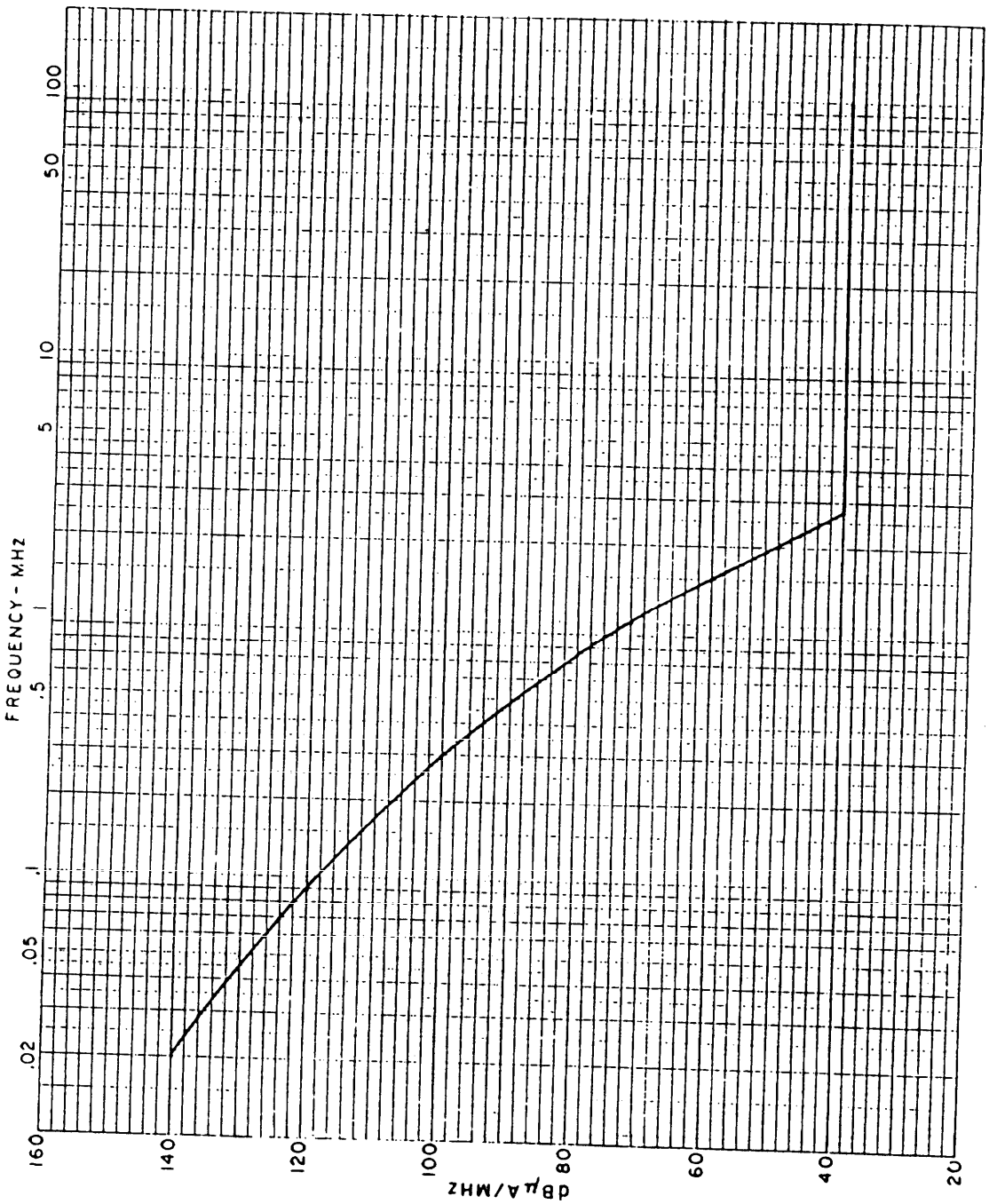


FIGURE 5. Broadband conducted emissions.

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Ⓑ Shock (specified pulse): MIL-STD-202, method 213, test condition F (1500 g's).

Vibration: 20 g's, 10 to 2,000 Hz.

Moisture resistance: Not applicable.

Electromagnetic interference: Maximum broadband conducted emission on power lines with 115 V ac, 400 Hz, 1.0 ampere resistive load during steady state and switching conditions (see figure 5).

Resistance to soldering heat: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041, MIL-STD-750.

PHYSICAL DATA:

Weight: 5 grams (typical).

Dimensions and configuration: See figure 1.

Terminal strength: 1 pound pull minimum.

Terminal solderability: Applicable.

Seal: Hermetic (solder or weld).

Minimum marking: Military part number, functional diagram and terminal identification, date code, and manufacturer's name or source code.

Ⓑ Part number: M28750/9-001 W, Y, or X to indicate the applicable screening level as specified in MIL-R-28750.

CONCLUDING MATERIAL

Ⓑ Custodians:
Navy - EC
Air Force - 85

Preparing activity:
Navy - EC

Agent:
DLA - EC

(Project 5945-0778-01)

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